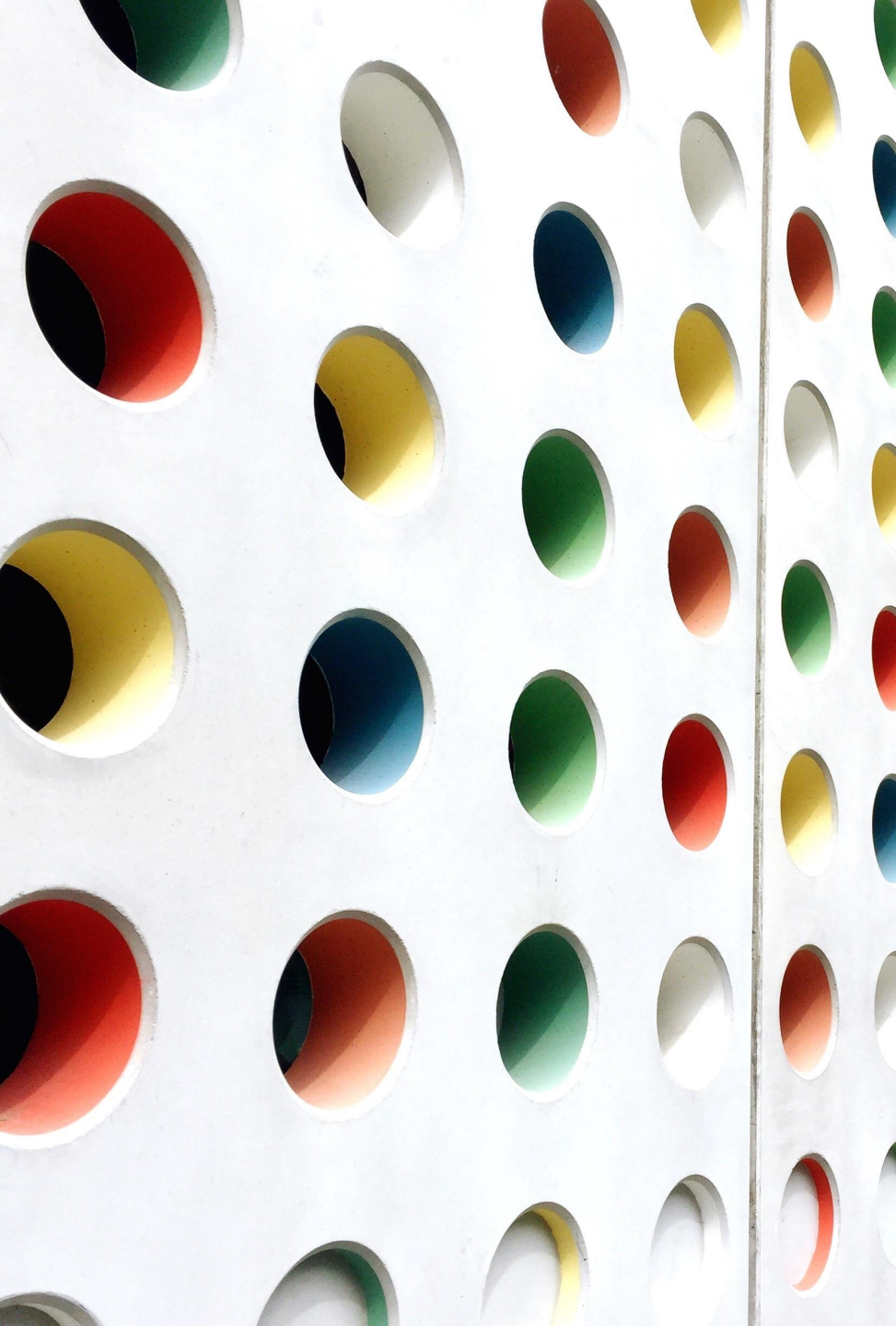


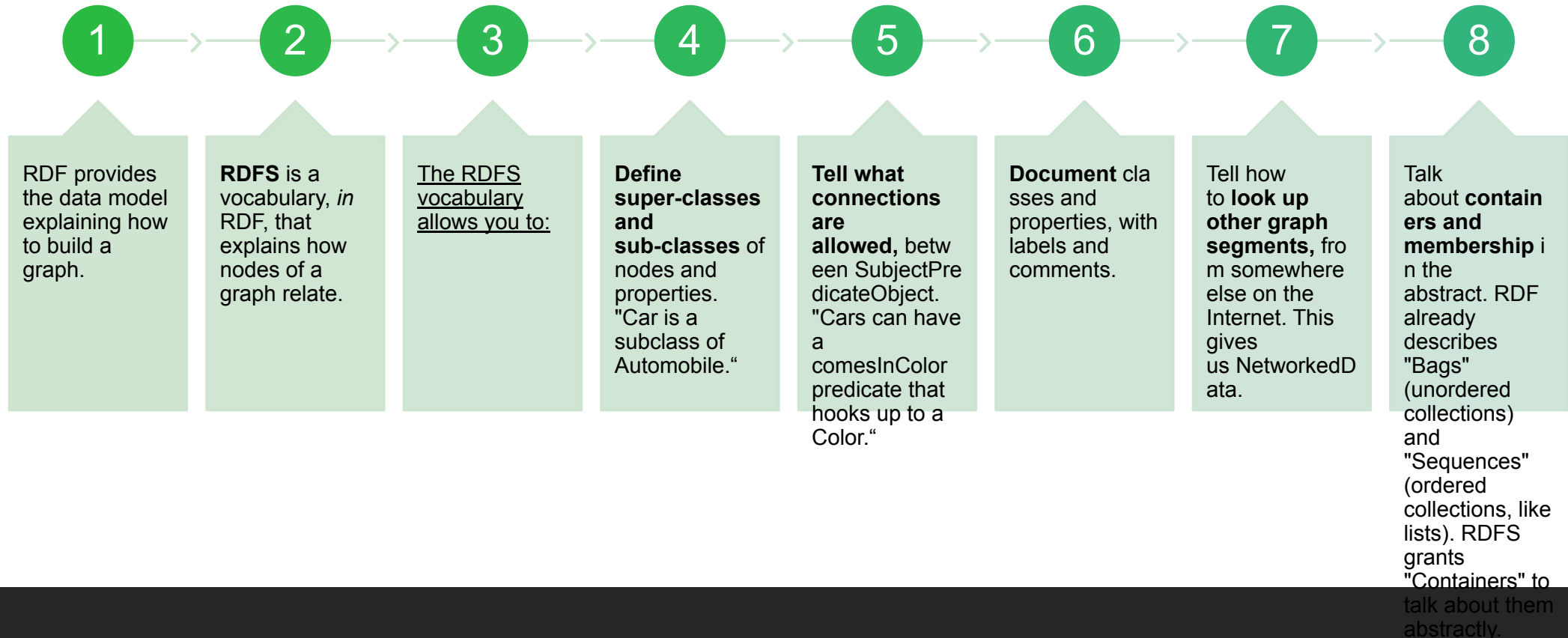


RDFS

BY

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Information:



Classes:

- **rdfs:Resource** is the class of everything. All things described by RDF are resources.
- **rdfs:Class** declares a resource as a class for other resources.

A typical example of `rdfs:Class` is `foaf:Person` in the Friend of a Friend (**FOAF**) vocabulary. An instance of `foaf:Person` is a resource that is linked to the class `foaf:Person` using the `rdf:type` **property**, such as in the following formal expression of the **natural-language** sentence: 'John is a Person'.

ex:John rdf:type foaf:Person

The definition of `rdfs:Class` is recursive: `rdfs:Class` is the class of classes, and so it is an instance of itself.

rdfs:Class rdf:type rdfs:Class

The other classes described by the RDF and RDFS specifications are:

- **rdfs:Literal** – literal values such as strings and integers. Property values such as textual strings are examples of RDF literals. Literals may be plain or typed.
- **rdfs:Datatype** – the class of datatypes. `rdfs:Datatype` is both an instance of and a subclass of `rdfs:Class`. Each instance of `rdfs:Datatype` is a subclass of `rdfs:Literal`.
- **rdf:XMLLiteral** – the class of XML literal values. `rdf:XMLLiteral` is an instance of `rdfs:Datatype` (and thus a subclass of `rdfs:Literal`).
- **rdf:Property** – the class of properties.

RDFS – Example with color patterns

```
<rdf:RDF
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"

  <rdfs:Class rdf:ID="lecturer">
    <rdfs:comment>
      The class of lecturers.
      All lecturers are academic staff members.
    </rdfs:comment>
    <rdfs:subClassOf rdf:resource="#academicStaffMember"/>
  </rdfs:Class>

  <rdfs:Class rdf:ID="academicStaffMember">
    <rdfs:comment>
      The class of academic staff members
    </rdfs:comment>
    <rdfs:subClassOf rdf:resource="#staffMember"/>
  </rdfs:Class>

  <rdfs:Class rdf:ID="staffMember">
    <rdfs:comment>
      The class of staff members
    </rdfs:comment>
  </rdfs:Class>

  <rdfs:Class rdf:ID="course">
    <rdfs:comment>
      The class of courses
    </rdfs:comment>
  </rdfs:Class>
```

```
<rdf:Property rdf:ID="involves">
  <rdfs:comment>
    It relates only courses to lecturers
  </rdfs:comment>
  <rdfs:domain rdf:resource="#course"/>
  <rdfs:range rdf:resource="#lecturer"/>
</rdf:Property>

<rdf:Property rdf:ID="isTaughtBy">
  <rdfs:comment>
    Inherits its domain ("course") and range ("lecturer")
    from its superproperty "involves"
  </rdfs:comment>
  <rdfs:subPropertyOf rdf:resource="#involves"/>
</rdf:Property>

<rdf:Property rdf:ID="phone">
  <rdfs:comment>
    It is a property of staff members,
    and takes literals as values.
  </rdfs:comment>
  <rdfs:domain rdf:resource="#staffMember"/>
  <rdfs:range rdf:resource="http://www.w3.org/2000/01/rdf-schema#Literal"/>
</rdf:Property>
```

DCMI Properties

Property	Definition
Contributor	An entity responsible for making contributions to the content of the resource
Coverage	The extent or scope of the content of the resource
Creator	An entity primarily responsible for making the content of the resource
Format	The physical or digital manifestation of the resource
Date	A date of an event in the lifecycle of the resource
Description	An account of the content of the resource
Identifier	An unambiguous reference to the resource within a given context
Language	A language of the intellectual content of the resource
Publisher	An entity responsible for making the resource available
Relation	A reference to a related resource
Rights	Information about rights held in and over the resource
Source	A Reference to a resource from which the present resource is derived
Subject	A topic of the content of the resource
Title	A name given to the resource
Type	The nature or genre of the content of the resource

RDF Tags with DC (Dublin Core)

```
<?xml version="1.0"?>
<rdf:RDF
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:dc="http://purl.org/dc/elements/1.1/">

  <rdf:Description rdf:about="https://www.w3schools.com">
    <dc:description>W3Schools - Free tutorials</dc:description>
    <dc:publisher>Refsnes Data as</dc:publisher>
    <dc:date>2008-09-01</dc:date>
    <dc:type>Web Development</dc:type>
    <dc:format>text/html</dc:format>
    <dc:language>en</dc:language>
  </rdf:Description>
</rdf:RDF>
```

RDF/RDFS Classes

Element	Class of	Subclass of
rdfs:Class	All classes	
rdfs:Datatype	Data types	Class
rdfs:Resource	All resources	Class
rdfs:Container	Containers	Resource
rdfs:Literal	Literal values (text and numbers)	Resource
rdf:List	Lists	Resource
rdf:Property	Properties	Resource
rdf:Statement	Statements	Resource
rdf:Alt	Containers of alternatives	Container
rdf:Bag	Unordered containers	Container
rdf:Seq	Ordered containers	Container
rdfs:ContainerMembershipProperty	Container membership properties	Property
rdf:XMLLiteral	XML literal values	Literal

RDF/RDFS Classes -- subclassOf

```
<?xml version="1.0"?>
```

```
<rdf:RDF  
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"  
  xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"  
  xml:base="http://www.animals.fake/animals#">
```

```
  <rdfs:Class rdf:ID="animal" />
```

```
  <rdfs:Class rdf:ID="horse">  
    <rdfs:subclassOf rdf:resource="#animal"/>  
  </rdfs:Class>
```

```
</rdf:RDF>
```


rdfs:domain	Property	Class	The domain of the resource
rdfs:range	Property	Class	The range of the resource
rdfs:subPropertyOf	Property	Property	The property is a sub property of a property
rdfs:subClassOf	Class	Class	The resource is a subclass of a class
rdfs:comment	Resource	Literal	The human readable description of the resource
rdfs:label	Resource	Literal	The human readable label (name) of the resource
rdfs:isDefinedBy	Resource	Resource	The definition of the resource
rdfs:seeAlso	Resource	Resource	The additional information about the resource
rdfs:member	Resource	Resource	The member of the resource
rdf:first	List	Resource	
rdf:rest	List	List	
rdf:subject	Statement	Resource	The subject of the resource in an RDF Statement
rdf:predicate	Statement	Resource	The predicate of the resource in an RDF Statement
rdf:object	Statement	Resource	The object of the resource in an RDF Statement
rdf:value	Resource	Resource	The property used for values

RDF/RDFS

Properties

RDF Attributes

Attribute	Description
rdf:about	Defines the resource being described
rdf:Description	Container for the description of a resource
rdf:resource	Defines a resource to identify a property
rdf:datatype	Defines the data type of an element
rdf:ID	Defines the ID of an element
rdf:li	Defines a list
rdf:_n	Defines a node
rdf:nodeID	Defines the ID of an element node
rdf:parseType	Defines how an element should be parsed
rdf:RDF	The root of an RDF document
xml:base	Defines the XML base
xml:lang	Defines the language of the element content

Model Sentence in RDF/XML coding

The students in course 6.001 are Amy, Tim, John, Mary, and Sue.

```
<rdf:RDF>
  <rdf:Description about="http://mycollege.edu/courses/6.001">
    <s:students>
      <rdf:Bag>
        <rdf:li resource="http://mycollege.edu/students/Amy"/>
        <rdf:li resource="http://mycollege.edu/students/Tim"/>
        <rdf:li resource="http://mycollege.edu/students/John"/>
        <rdf:li resource="http://mycollege.edu/students/Mary"/>
        <rdf:li resource="http://mycollege.edu/students/Sue"/>
      </rdf:Bag>
    </s:students>
  </rdf:Description>
</rdf:RDF>
```

Use of Container and Bag

```
<rdf:Bag ID="pages">
  <rdf:li resource="http://foo.org/foo.html" />
  <rdf:li resource="http://bar.org/bar.html" />
</rdf:Bag>

<rdf:Description about="#pages">
  <s:Creator>Ora Lassila</s:Creator>
</rdf:Description>
```

Collection – [sequence, bag, alternative]

```
<rdf:RDF
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:dc="http://purl.org/metadata/dublin_core#"

  <rdf:Description about="http://www.foo.com/cool.html">
    <dc:Creator>
      <rdf:Seq ID="CreatorsAlphabeticalBySurname">
        <rdf:li>Mary Andrew</rdf:li>
        <rdf:li>Jacky Crystal</rdf:li>
      </rdf:Seq>
    </dc:Creator>

    <dc:Identifier>
      <rdf:Bag ID="MirroredSites">
        <rdf:li rdf:resource="http://www.foo.com.au/cool.html"/>
        <rdf:li rdf:resource="http://www.foo.com.it/cool.html"/>
      </rdf:Bag>
    </dc:Identifier>

    <dc:Title>
      <rdf:Alt>
        <rdf:li xml:lang="en">The Coolest Web Page</rdf:li>
        <rdf:li xml:lang="it">Il Pagio di Web Fuba</rdf:li>
      </rdf:Alt>
    </dc:Title>
  </rdf:Description>
</rdf:RDF>
```


Dublin Core Metadata

```
<rdf:RDF
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:dc="http://purl.org/metadata/dublin_core#">

  <rdf:Description about="http://www.dlib.org">
    <dc:Title>D-Lib Program - Research in Digital Libraries</dc:Title>
    <dc:Description>The D-Lib program supports the community of people
      with research interests in digital libraries and electronic
      publishing.</dc:Description>
    <dc:Publisher>Corporation For National Research Initiatives
  </dc:Publisher>
    <dc:Date>1995-01-07</dc:Date>
    <dc:Subject>
      <rdf:Bag>
        <rdf:li>Research; statistical methods</rdf:li>
        <rdf:li>Education, research, related topics</rdf:li>
        <rdf:li>Library use Studies</rdf:li>
      </rdf:Bag>
    </dc:Subject>
    <dc:Type>World Wide Web Home Page</dc:Type>
    <dc:Format>text/html</dc:Format>
    <dc:Language>en</dc:Language>
  </rdf:Description>
</rdf:RDF>
```

Dublin Core Metadata

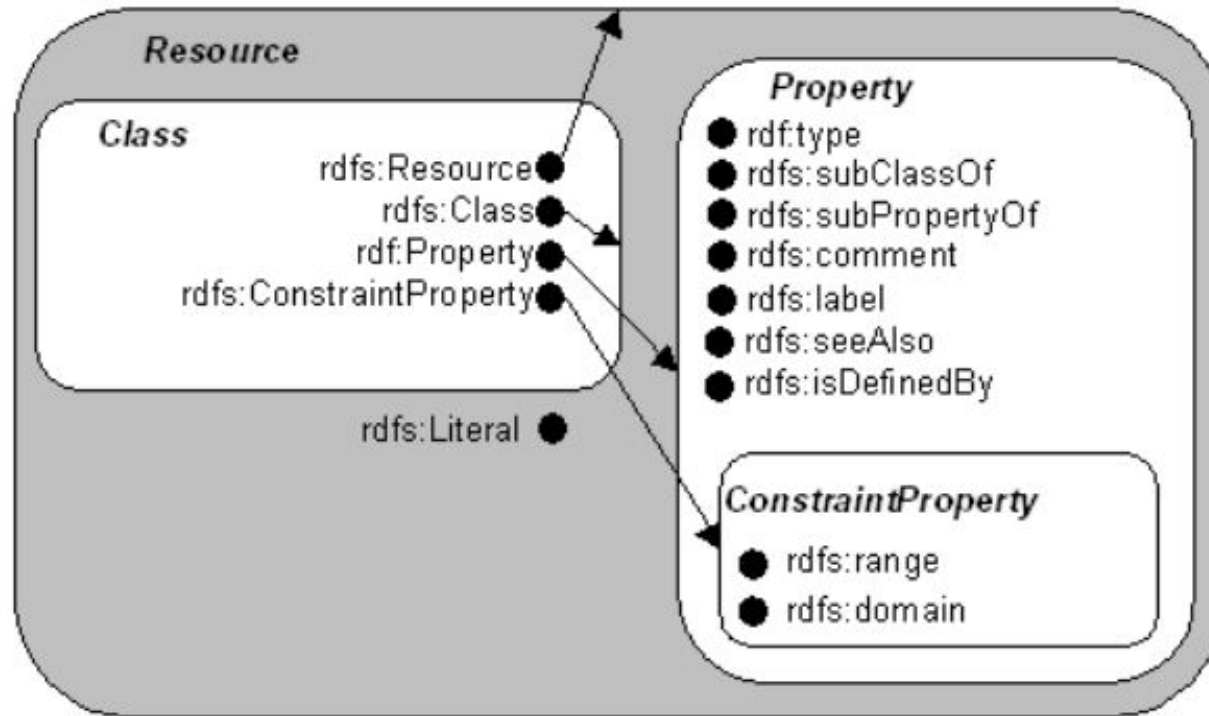
```
<rdf:RDF
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:dc="http://purl.org/metadata/dublin_core#"
  xmlns:dcq="http://purl.org/metadata/dublin_core_qualifiers#">

  <rdf:Description about="http://www.dlib.org/dlib/may98/05contents.html">
    <dc>Title>DLIB Magazine - The Magazine for Digital Library Research
      - May 1998</dc>Title>
    <dc:Description>D-LIB magazine is a monthly compilation of
      contributed stories, commentary, and briefings.</dc:Description>
    <dc:Contributor rdf:parseType="Resource">
      <dcq:AgentType
        rdf:resource="http://purl.org/metadata/dublin_core_qualifiers#Editor"/>
      <rdf:value>Amy Friedlander</rdf:value>
    </dc:Contributor>
    <dc:Publisher>Corporation for National Research
      Initiatives</dc:Publisher>
    <dc>Date>1998-01-05</dc>Date>
    <dc:Type>electronic journal</dc:Type>
    <dc:Subject>
      <rdf:Bag>
        <rdf:li>library use studies</rdf:li>
        <rdf:li>magazines and newspapers</rdf:li>
      </rdf:Bag>
    </dc:Subject>
    <dc:Format>text/html</dc:Format>
    <dc:Identifier>urn:issn:1082-9873</dc:Identifier>
    <dc:Relation rdf:parseType="Resource">
      <dcq:RelationType
        rdf:resource="http://purl.org/metadata/dublin_core_qualifiers#IsPartOf"
      >
      <rdf:value resource="http://www.dlib.org"/>
    </dc:Relation>
  </rdf:Description>
</rdf:RDF>
```

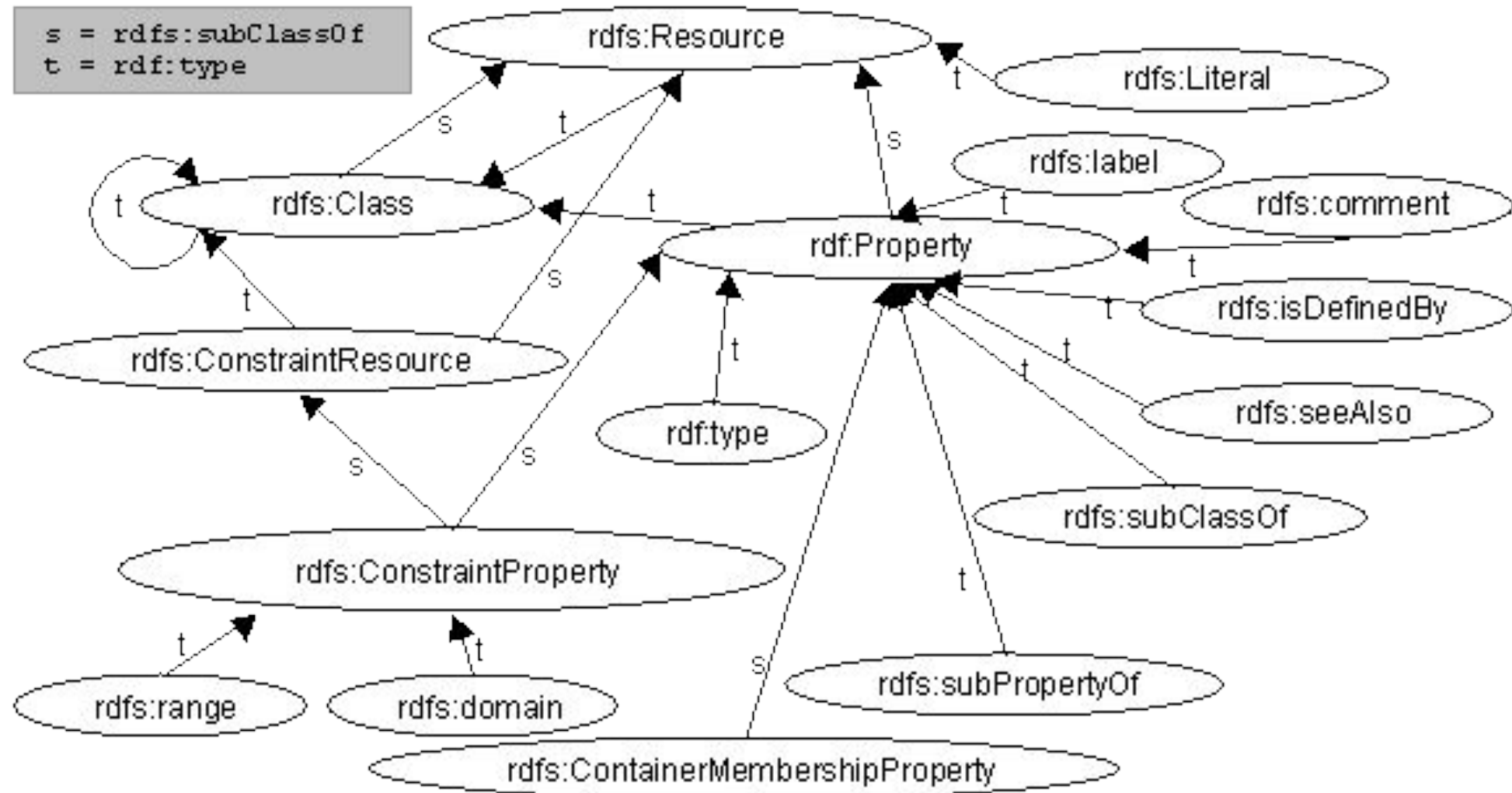
RDF property values with XML attributes

The RDF abbreviated syntax can frequently be used to write property values that are strings in XML attribute form

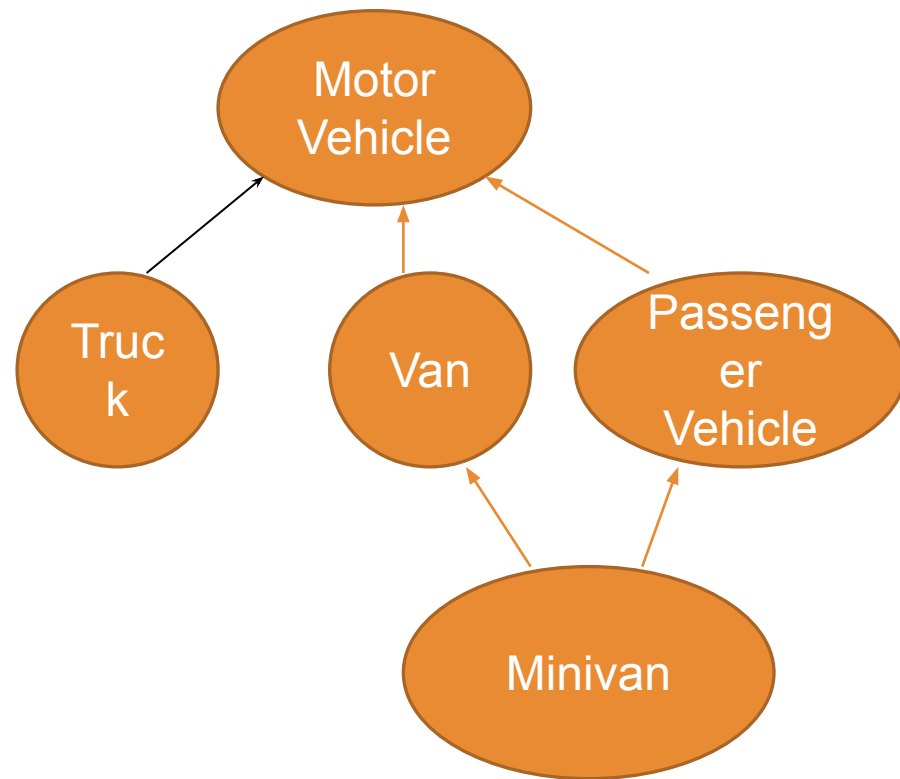
```
<rdf:RDF
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:dc="http://purl.org/metadata/dublin_core#"
  <rdf:Description about="http://www.foo.com/cool.html">
    <dc:Creator>
      <rdf:Seq ID="CreatorsAlphabeticalBySurname"
        rdf:1="Mary Andrew"
        rdf:2="Jacky Crystal"/>
    </dc:Creator>
  </rdf:Description>
</rdf:RDF>
```



RDFS—Classes and Resources as Sets and Elements



We first define a class 'MotorVehicle'. We then define 3 subclasses of 'MotorVehicle', namely 'PassengerVehicle', 'Truck' and 'Van'. We then define a class 'Minivan' which is a sub-class of both 'Van' and 'PassengerVehicle'.



```
<rdf:RDF xml:lang="en"
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:rdfs="http://www.w3.org/TR/1999/PR-rdf-schema-19990303#">

  <!-- Note: this RDF schema would typically be used in RDF instance data
  by referencing it with an XML namespace declaration, for example
  xmlns:xyz="http://w3.org/examples/vehicles#". This allows us to use
  abbreviations such as xyz:MotorVehicle to refer unambiguously to the
  RDF class 'MotorVehicle'. -->

  <rdf:Description ID="MotorVehicle">
    <rdf:type resource="http://www.w3.org/TR/1999/PR-rdf-schema-19990303#Class"/>
    <rdfs:subClassOf
      rdf:resource="http://www.w3.org/TR/1999/PR-rdf-schema-19990303#Resource"/>
  </rdf:Description>

  <rdf:Description ID="PassengerVehicle">
    <rdf:type resource="http://www.w3.org/TR/1999/PR-rdf-schema-19990303#Class"/>
    <rdfs:subClassOf rdf:resource="#MotorVehicle"/>
  </rdf:Description>

  <rdf:Description ID="Truck">
    <rdf:type resource="http://www.w3.org/TR/1999/PR-rdf-schema-19990303#Class"/>
    <rdfs:subClassOf rdf:resource="#MotorVehicle"/>
  </rdf:Description>

  <rdf:Description ID="Van">
    <rdf:type resource="http://www.w3.org/TR/1999/PR-rdf-schema-19990303#Class"/>
    <rdfs:subClassOf rdf:resource="#MotorVehicle"/>
  </rdf:Description>

  <rdf:Description ID="MiniVan">
    <rdf:type resource="http://www.w3.org/TR/1999/PR-rdf-schema-19990303#Class"/>
    <rdfs:subClassOf rdf:resource="#Van"/>
    <rdfs:subClassOf rdf:resource="#PassengerVehicle"/>
  </rdf:Description>

</rdf:RDF>
```

If the property
biologicalFather is a
sub-property of the broader
property biologicalParent,
and if Fred is the
biologicalFather of John,
then it is implied that Fred
is also the biologicalParent
of John.

Fred is BiologicalFather of John.

BiologicalFather isSubpropertyOf BiologicalParent.

So so so

Fred is BiologicalParent of John.

```
<rdf:RDF xml:lang="en"
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:rdfs="http://www.w3.org/TR/1999/PR-rdf-schema-19990303#">

  <rdf:Description ID="biologicalParent">
    <rdf:type resource="http://www.w3.org/1999/02/22-rdf-syntax-ns#Property"/>
  </rdf:Description>

  <rdf:Description ID="biologicalFather">
    <rdf:type resource="http://www.w3.org/1999/02/22-rdf-syntax-ns#Property"/>
    <rdfs:subPropertyOf rdf:resource="#biologicalParent"/>
  </rdf:Description>
</rdf:RDF>
```

For more example watch the linked file

[C:\Users\admin\Desktop\RDFS Example.docx](#)

Task to do for RDF/ RDFS

Make digital library environment having 10 books detail of 4 different majors with name of book, author, year, title and publisher details using bag, container, sequence, literal, alternative, class, datatype, comments. You can use vocab. of FOAF/DC